Wireless Technologies

The scope of application or usage of an IoT network has a significant impact on the choice of the optimal solution in terms of adequate hardware and software infrastructure. The wide scope of IoT usage, where each of them specifically dictates hardware, software and communications needs, defines the factors that must be captured by the analysis process. The choice of a particular technology depends on the scope of the IoT network because it can significantly affect the design process.

A number of features that have a significant impact on the process of designing and implementing an IoT network include:

- Reach
- Bandwidth
- Coverage
- Power consumption
- Reliability
- Latency
- Cost-effectiveness

Range

The application of IoT in smart city areas on the one hand, as well as smart meter readings on the other, leads to conflicting requirements regarding the reach of a communications network and can extend from a few kilometers to just a few centimeters of distance. The example above explains the importance of range as an important criterion for choosing the right IoT network implementation. Depending on the distance of IoT nodes, communication networks are divided into three basic categories:

- short-range networks
- medium-range networks
- and long-range network

Bandwidth

Depending on the application of the IoT network, nodes may send information periodically, e.g. in smart home systems, fire detection sensor connects to the network and sends information in case smoke/fire is detected. In contrast, if IoT nodes are used in security applications such as security cameras, then there must be constant communication and the transmission of significant amounts of data. Increasing bandwidth affects the reach and power consumption, which entails other factors such as the cost and cost-effectiveness of the entire system, making bandwidth an important criterion for making the right decision when choosing an IoT network implementation.

Reliability

If the designed IoT system is used in medical systems to monitor the patient's vital parameters, one of the most important features is that the correct information is successfully transmitted to the destination as soon as possible. On the other hand, in the case of the application of the IoT system in agriculture, the transmission or omission of transmission of information on e.g. soil humidity at a certain interval, will not cause significant damage. Increasing reliability also has a significant impact on increasing bandwidth and

cost, which puts reliability at the center of an important criterion for choosing an IoT network implementation

Latency

Network latency is defined as the time it takes for a request to be forwarded from the transmitting device to the receiving device, including the time it takes the destination device to process it. In cases of application of IoT nodes in industry, sending control information and signals, e.g. for the management of the power grid, it can be concluded that in this case, it is very important that the information is transmitted to the destination so, if a problem occurs, then the network can be switched off as soon as possible. In contrast, the application of the IoT network in agriculture, e.g. soil temperature measurements, the transmission time can be considered negligible. Reduced latency has the effect of reducing the lifespan of autonomous IoT nodes, and is, therefore, an essential factor in the correct choice of IoT network implementation

Energy efficiency

Energy efficiency, that is, the energy consumption of IoT nodes can certainly be seen as one of the key parameters when designing an IoT network. The concept of IoT itself encompasses autonomous sensor nodes that have an increased lifespan, which is a considerable problem in the development process. By introducing state-of-the-art technological solutions into the process of developing hardware components of IoT, the problem of autonomous power sources has been significantly resolved. However, while the capacity and quality of autonomous power sources are increased, energy efficiency remains a key element in the design process. Some of the reasons that lead to increased consumption are the constant activity of IoT nodes and their sensor units, as well as the large exchange of data within the sensor network. Minimizing the activity time of IoT nodes, eliminating full-duplex or half-duplex communication models, and reducing data transmission time as well as their amount (encryption application and hexadecimal format transfer) have made significant savings in terms of energy efficiency. Also, when designing IoT nodes, it is necessary to pay attention to the consumption of sensor elements that can significantly cut the life of an IoT node. The configuration of nodes in the network, as well as the atmospheric conditions of the environment in which the IoT nodes are located, can also affect energy consumption.

Based on the above, it can be concluded that low power consumption in wireless IoT networks is a feature that is possible and feasible, but to a considerable extent limited

Coverage

Wireless coverage is defined as the extent of the area to which the wireless signals are transmitted, and generally can be divided into five categories that differ in distance and speed of access

- Wireless Body Area Network (WBAN) - used to monitor and collect vital parameters of a person,
- Wireless Personal Area Network (WPAN) - Provide short-range communication for devices.
- Wireless Local Area Network (WLAN) - provide communication in a restricted area.
- Wireless Metropolitan Area Network (WMAN) - networks that provide communications at the city level.
- Wireless Wide Area Network (WWAN) - connect large geographical areas (countries or continents).

The challenge of picking the right communication technology for the implementation of the IoT network depends largely on the application of the designed IoT system. For instance, the application of IoT in designing smart cities, or smart agriculture, coverage differs significantly from the application of IoT for medical purposes. In these cases, coverage can range from a few centimeters, up to several tens of kilometers. It is important to note that greater network coverage does not necessarily mean better data flow or speed of exchange. Coverage is a necessary but not sufficient condition for data transmission and sharing to be possible

Cost-effectiveness

Cost-effectiveness is an aggregate criterion that is closely related to the business model and is proportionally dictated by the budget, needs, and capabilities that the designed IoT system must have. Cost-effectiveness is calculated according to all other criteria, e.g. systems where large amounts of data are transmitted over the network, depending on the topology of the network nodes (if the nodes are at a shorter distance from the system's data concentrator and close to the power source), Wi-Fi could be the best choice. If it is a WAN system, then one network technology that works on the principle of mobile data will be an adequate choice. The selection of the appropriate network should primarily comply with the basic requirements of the system to select the network that will have optimal cost-effectiveness for a particular IoT system.

Technologies

Wireless communication technologies are defined as the ability to transmit data or information wirelessly from one device to another, and as such are an integral part of any IoT system. Depending on the domain of application of the IoT system, information is transmitted from one device to another at different distances that can range from a few centimeters to several hundred kilometers through clearly defined transmission channels.

Due to the heterogeneity of the market, as well as the technologies available today, there is no unique technology for wireless transmission, and most often they can be divided by the length of the electromagnetic waves and the frequencies at which they operate. Figure 1 and Table 1 below depict an IOT general architecture and the comparative statistics respectively.

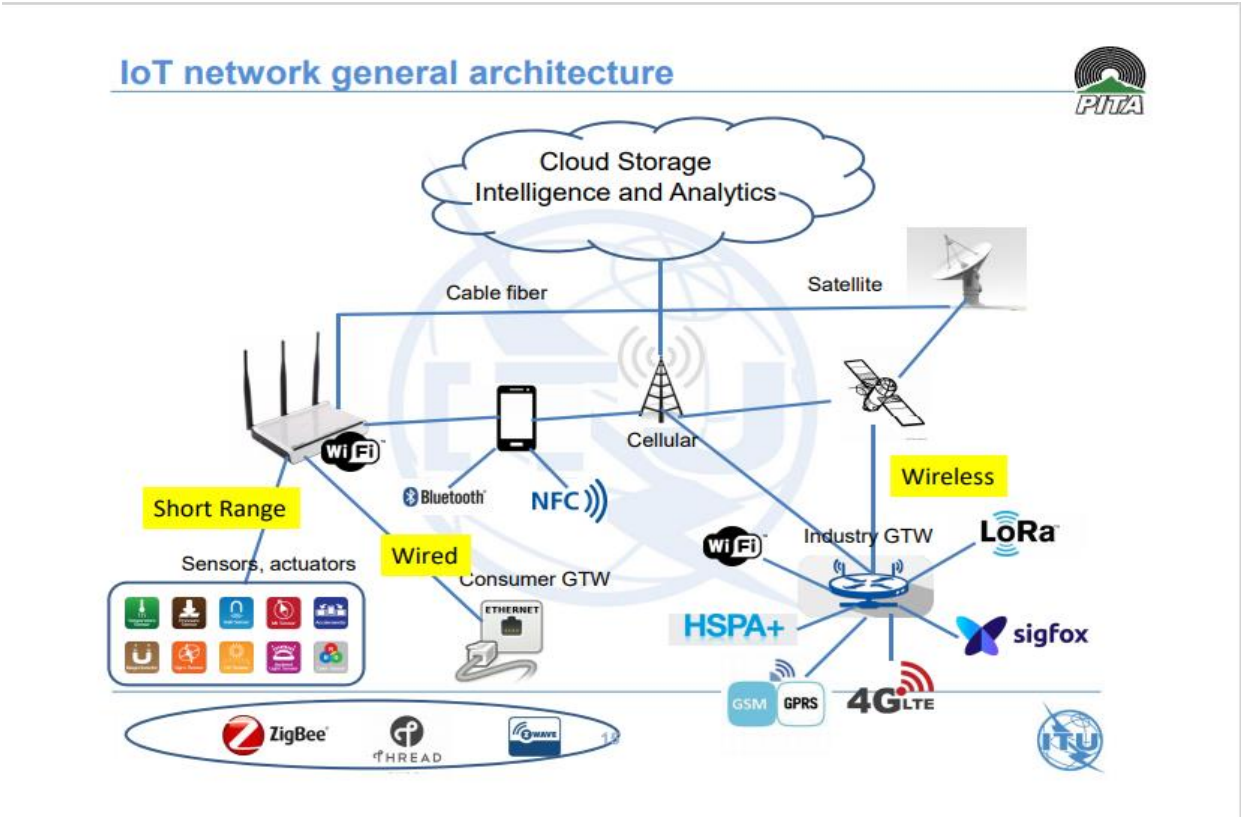


Figure 1

TABLE I. COMPARATION OF WIRELESS COMMUNICATION TECHNOLOGIES

Wireless Communication Technologies	Parameters							
	Range	*Bandwidth*	*Reliability*	*Latency*	*Power use*	*Coverage*	*Cost per module*	*Example of use*
Bluetooth Low Energy	< 10 m	< 2.4 GHz	Medium	< 10 ms	Low	Small	$2 – $5	Blood pressure measurement
Near-field Communication (NFC)	< 4 cm	13.56 MHz	High	100 ~ 250 ms	Very Low	Very small	Reader $5 - $40 Tag < $1	Mobile payments
ZigBee	< 100 m	< 2.4 GHz	High	< 20 ms	Low	Small	$1 - $5	Ground parking sensors
Wi-Fi	< 100 m	2.4 GHz – 5 GHz	High	< 10 ms	Medium	Small	$1 - $10	Remote control robot car
LoRaWAN	5 km urban 20 km rural	500 KHz – 900 MHz	Medium	1 ~ 10 s	Low	Large	$2 – $25	Smart trash bins
Narrow-band IoT (NB-IoT)	> 10 km	450 MHz-3.5 GHz	High	1.5 ~ 10 s	Low	High	$5 – $30	Smart warehouse
SigFox	10 km urban 50 km rural	100 Hz – 900 MHz	Medium	1 ~ 30 s	Low	Large	$4 – $60	Smart wineyards
3g/4g/LTE	> 30 km	200 KHz – 900 MHz	High	100 ms ~ 1 s	Low-High	Large	$4 – $35	Forest fire detection

Some of the common technologies used currently are described below.

A. Bluetooth

Bluetooth is a type of wireless connection developed by the Bluetooth Special Interest Group (SIG). The basic purpose is communication between devices at short distances and is most commonly used to connect a computer peripheral. It operates in the 2.4 GHz ISM (industrial, scientific and medical) band, and more recently it is the de-facto standard in the mobile industry. Its characteristics are summarized in figure 2 below.

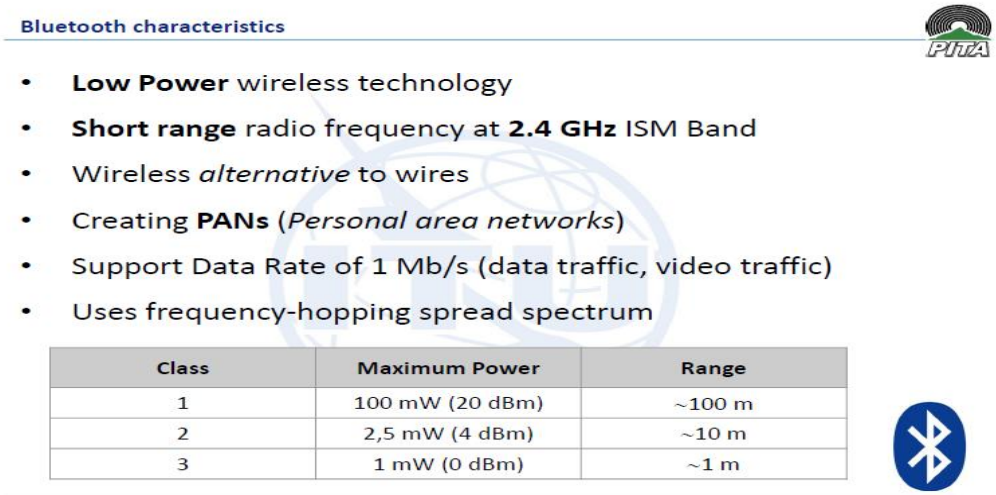


Figure 2

Bluetooth Low Energy (sometimes referred to as Bluetooth Smart) is a variant that was launched in 2011 as part of the Bluetooth 4.0 specification, and the advantage over earlier versions is that it uses less power, has less setup time, and supports an unlimited star type topology by the number of nodes. BLE is specific for devices with low bandwidth, as well as for devices powered by autonomous power sources. BLE characteristics are summarized in figure 3.

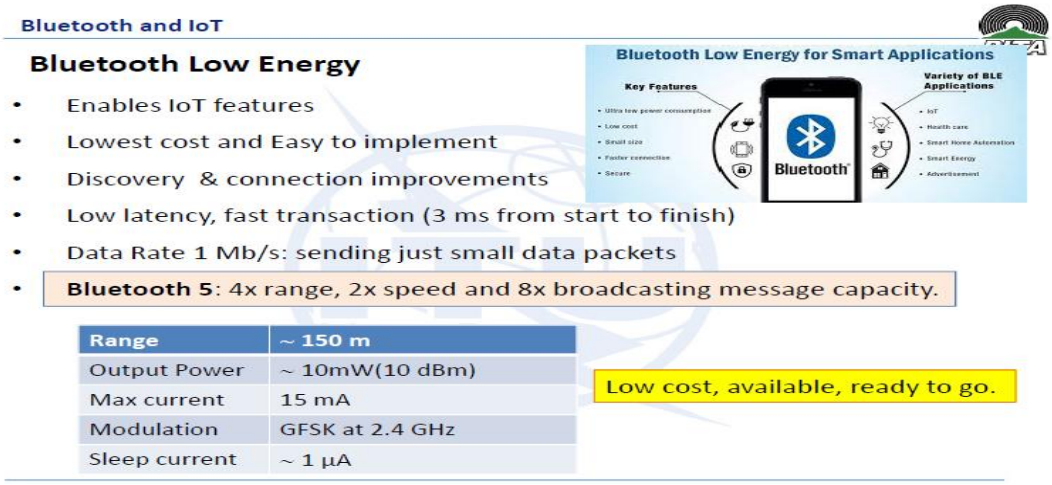


Figure 3.

B. Near-field Communication (NFC)

NFC is a wireless communications technology that was developed by Sony and Philips. It works on the principle of magnetic coupling between the devices and enables two-way data transfer when the two devices are placed side by side (up to a maximum distance of 10 cm). NFC is RFID based and operates at up to 13.56 MHz. Includes a tag that contains a small amount of data, and can be read-only or read-write when the device rewrites the data. The most common application is in mobile devices for contactless payment, that is, storage and access to all types of personal data

C. Wi-Fi

Represents the most commonly used communication standard, assigned as 802.11 by the Institute of Electrical and Electronics Engineers (IEEE). The technology is promoted by the non-profit Wi-Fi Alliance, which certifies products with WiFi interoperability. It works in the 2.4 to 5 GHz range, and an access point and a wireless network adapter are required to set up a core network of this type. Wi-Fi in its reach, data transfer speed, and market penetration is a key technology available in the market today. Figure 4 depicts the characteristics of Wi-Fi variants.

WiFi

- Wireless technology
- Alternative to Wired Technologies
- IEEE 802.11 standard for WLANs

Standard	Frequency bands	Throughput	Range
WiFi a (802.11a)	5 GHz	54 Mbit/s	10 m
WiFi B (802.11b)	2.4 GHz	11 Mbit/s	140 m
WiFi G (802.11g)	2.4 GHz	54 Mbit/s	140 m
WiFi N (802.11n)	2.4 GHz / 5 GHz	450 Mbit/s	250 m
IEEE 802.11ah	900 MHz	8 Mbit/s	100 M

Figure 4

Wi-Fi is used widely with different devices as depicted in figure 5.

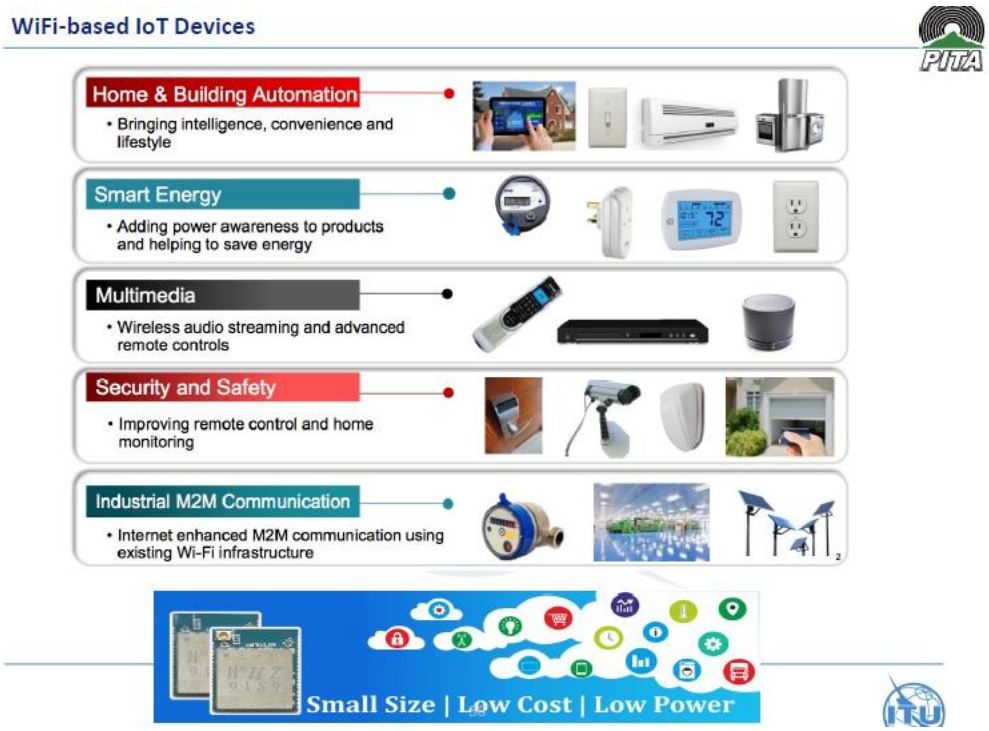


Figure 5: Wi-Fi devices

D. Cellular 3G/4G/5G LTE

The cellular network is a large area covered by radio waves that are divided into defined shapes and sizes called cells. Each area has one transceiver (base station), which connects one device to another in the same or different cell. Such networks offer low energy consumption and a large coverage area. The cellular network first appeared in the 1990s under the designation 1G (First generation). Since then, up to the present day, new versions (2G, 3G, 4G) are emerging, as well as inter-versions, which differ in characteristics, primarily by transmission mode (analog-digital) and power consumption. 5G and LTE (Long term Evolution) represent the latest generation of cellular technologies and are popular with mobile phones and IoT devices that require high bandwidth. 5G networks, compared to the previous generation, provide more speed and more channels, which results in increased capacity and allows more devices to connect.

E. LPWAN – Low Power Wide Area Networks

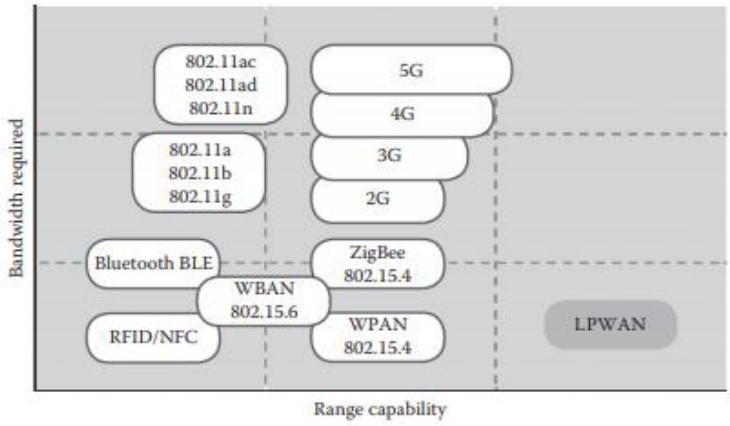
Low power wide area networking technologies (LPWAN) technologies are used extensively for communication in the IoT ecosystem. LPWAN technologies are superior when compared to Bluetooth and BLE for M2M communication because of their cost effectiveness and low-power consumption. LPWAN technology is ideal for connecting devices that send small amounts of data over a long range with battery efficiency. An ideal example of devices that fall under this category are sensors, which are used for transmitting data within smart homes, buildings, parking systems, etc.

The key features of LPWAN that make it suitable for IoT ecosystem are the following:

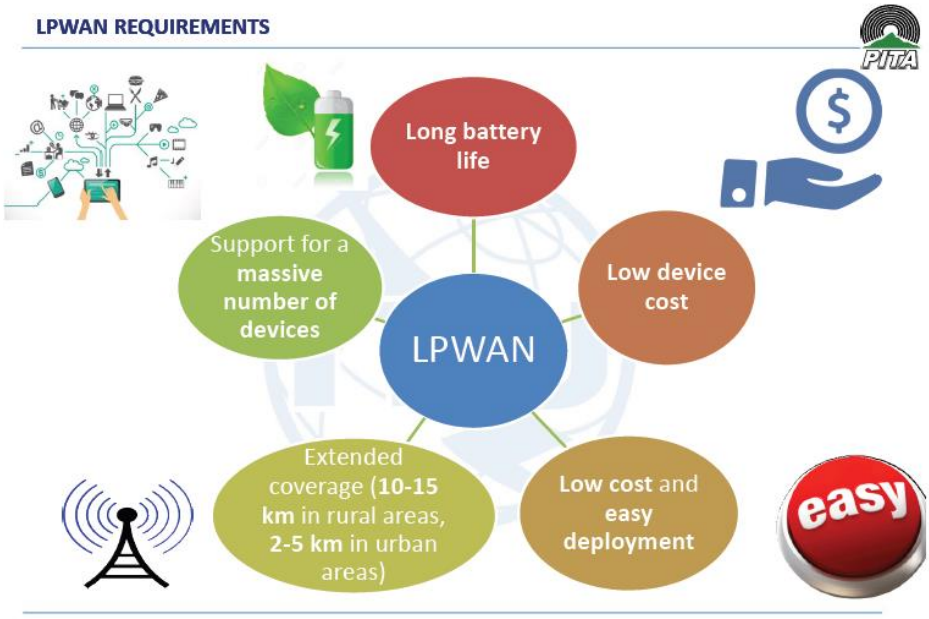
- Long-range communication: Ability to support nodes that are greater than or equal to 10 km distance from the gateway. However, the correct distance is based on the LPWAN technology that is used.
- Low transmission data rate: Less than 5000 bits of data are sent per second. Often only 20–256 bytes per message sent several times a day.
- Low-power consumption: This provides very long battery life for the devices. Many times, the battery life may last up to 10 years.

LPWAN technology is ideally suited for the following two types of applications:

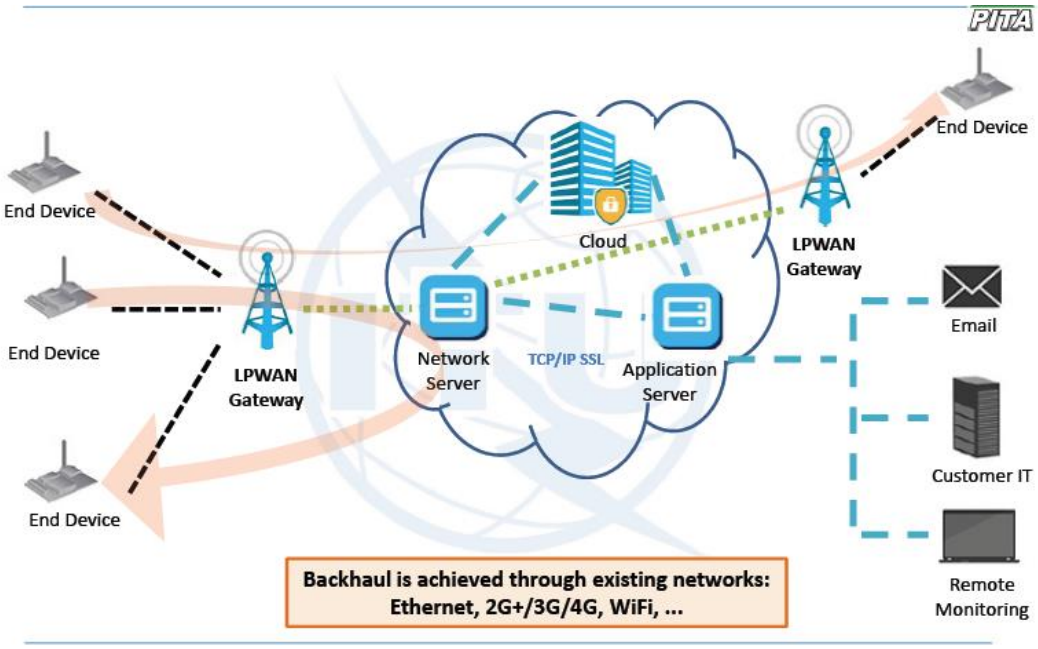
1. Fixed, medium-to-high density connections: This is mainly used in cities and for buildings as an alternative option for cellular communications. Some common examples are smart grids, GPS-based asset tracking systems, and smart lighting systems.
2. Long life, battery powered applications: For applications that need a long range. Some exam ples are water meters, gas detectors, smart agriculture systems, etc.



Bandwidth requirements of LPWAN technologies



LPWAN general Requirements



General architecture of LPWAN

F. SigFox

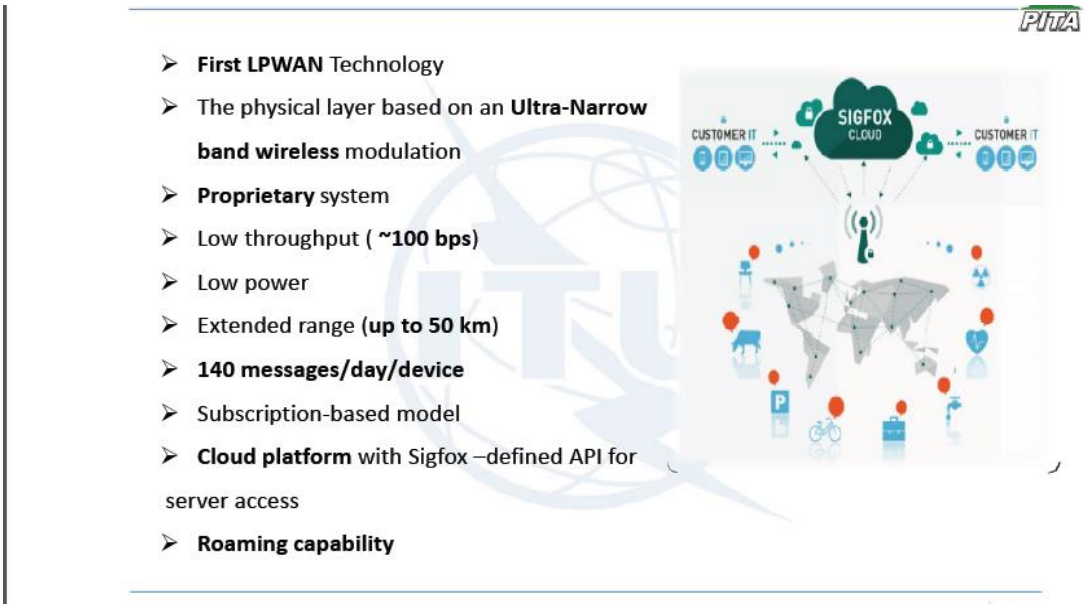
SigFox LP-WAN is an end-to-end system consisting of the presence of a certified modem and ending with a web-based application. Developers must acquire a modem from a certified manufacturer to integrate into their IoT endnode device. Alternatively, there are third-party service providers that make SigFox-compatible access point networks available to handle traffic between the endnodes and SigFox servers. The SigFox servers manage the endnode devices in the network, collect their data traffic, and then make the data and other information available to the users through a web-based API.

To provide security, the SigFox system uses frequency hopping, which mitigates the risks of message interception and channel blocking. Furthermore, SigFox provides anti-replay mechanisms in their servers to avoid replay attacks. The content and format of data sent in the transmission is user-defined so only the user knows how to interpret their device data.

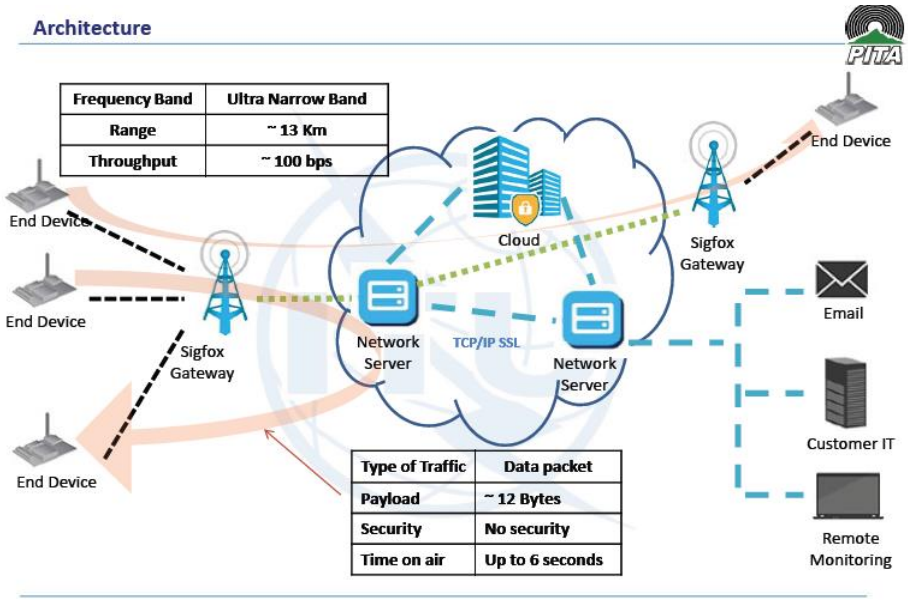
Frequency band: 868MHz/902MHz ISM

Channel width: Ultra narrow band

Range: 30-50km (rural), 3-10km (urban),



Sigfox Overview



Sigfox Architecture

G. LoRaWAN

The LoRaWAN architecture is a "star of stars" structure of endnode devices connecting through gateways in order to connect to network servers. The way LoRaWAN works is that the wireless hop between the endnodes and gateway uses a Semtech proprietary spread spectrum radio scheme, which they call a *chirp*. The endnodes communicate by chirping the gateway when they have data to transmit. The radio scheme allows the network server to manage the data rate for each connected device, which allows the LoRaWAN connections to decide between payload and range dependent on local radio conditions.

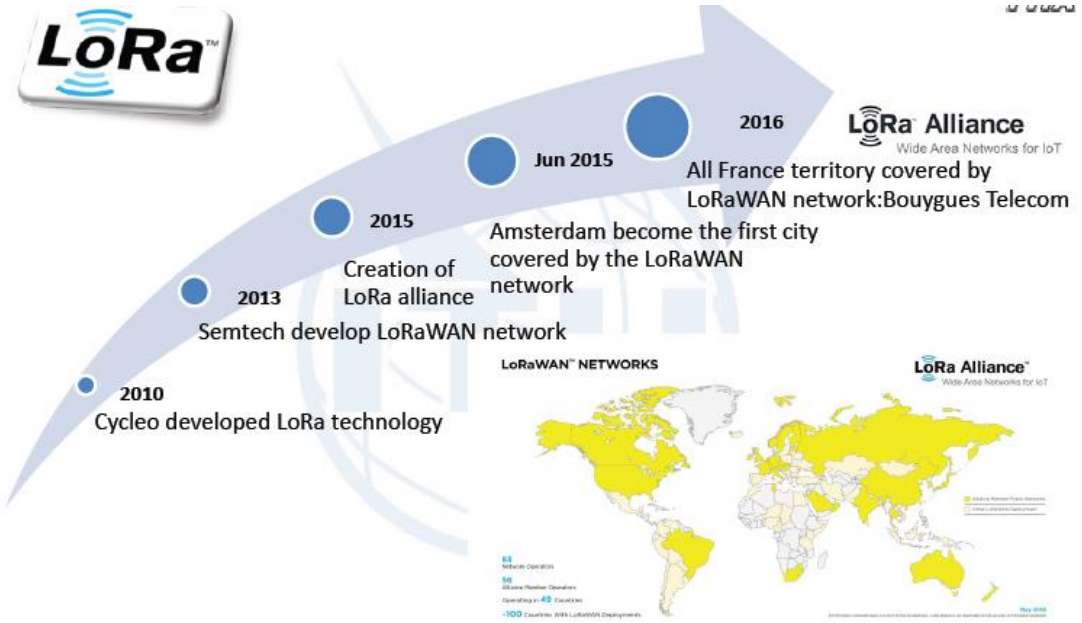
The LoRaWAN network topology has three classes of endnode device. Class A are bidirectional devices, which have a scheduled uplink transmission window and two, short downlink receive windows. Class B devices have additional, scheduled downlink windows and Class C devices have open receive windows. Security for LoRaWAN includes the use of unique networks and device-encryption keys.

Frequency band: 433/868/780/915MHz ISM,

Channel width EU: 8x125kHz, US 64x125kHz/8x125kHz

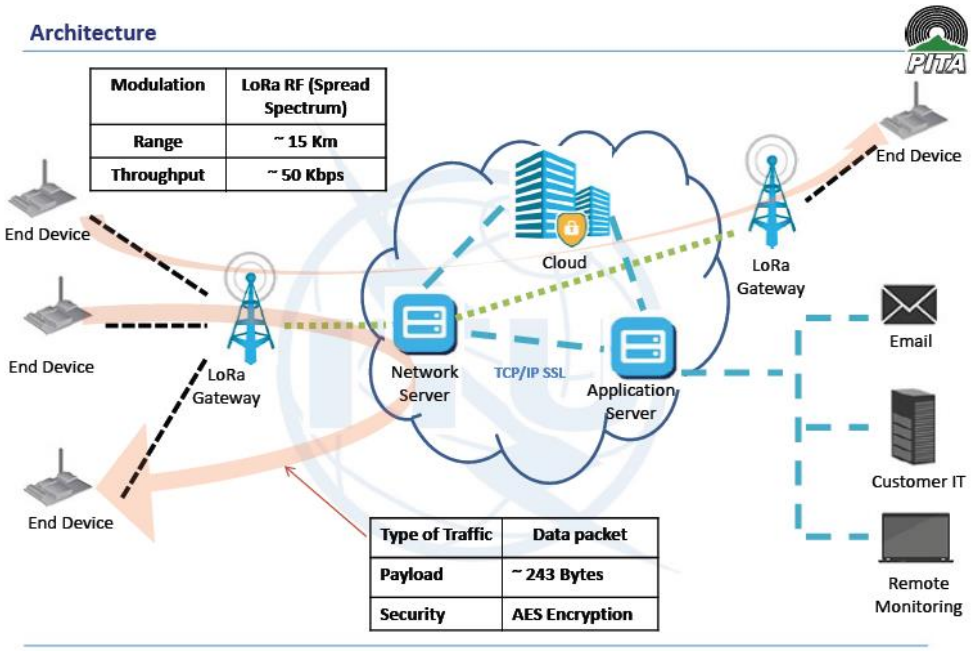
Modulation: Chirp Spread Spectrum

Range: 2-5k (urban), 15k (rural)

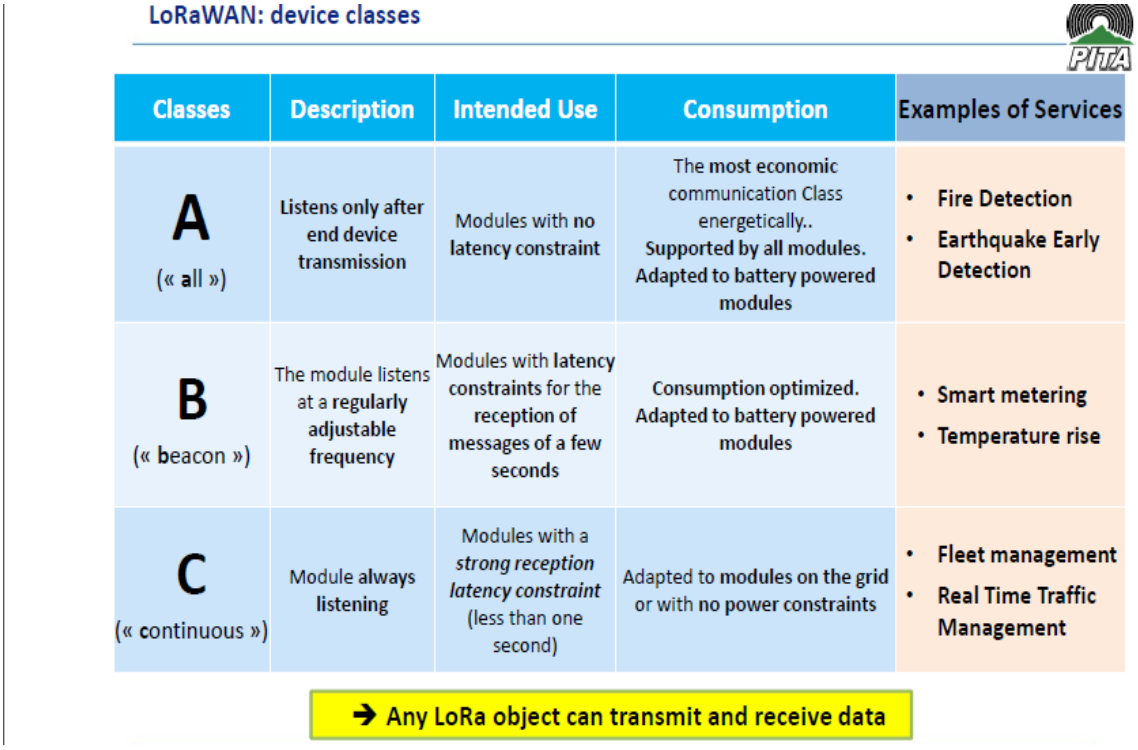


LoRa roadmap

- LoRaWAN is a Low Power Wide Area Network
- LoRa modulation: a version of Chirp Spread Spectrum (CSS) with a typical channel bandwidth of 125KHz
- High Sensitivity (End Nodes: Up to -137 dBm, Gateways: up to -142 dBm)
- Long range communication (up to 15 Km)
- Strong indoor penetration: With High Spreading Factor, Up to 20dB penetration (deep indoor)
- Occupies the entire bandwidth of the channel to broadcast a signal, making it robust to channel noise.
- Resistant to Doppler effect, multi-path and signal weakening.



LoRa Architecture



LoRaWAN: device classes

Classes	Description	Intended Use	Consumption	Examples of Services
A (« all »)	Listens only after end device transmission	Modules with no latency constraint	The most economic communication Class energetically.. Supported by all modules. Adapted to battery powered modules	• Fire Detection • Earthquake Early Detection
B (« beacon »)	The module listens at a regularly adjustable frequency	Modules with latency constraints for the reception of messages of a few seconds	Consumption optimized. Adapted to battery powered modules	• Smart metering • Temperature rise
C (« continuous »)	Module always listening	Modules with a *strong reception latency constraint* (less than one second)	Adapted to modules on the grid or with no power constraints	• Fleet management • Real Time Traffic Management

→ Any LoRa object can transmit and receive data

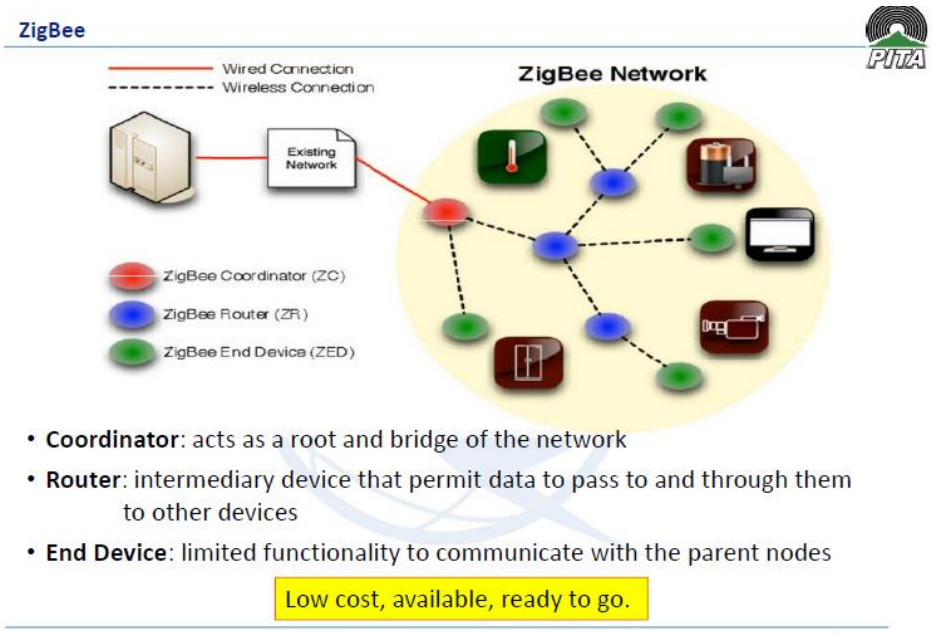
LoRaWAN device classes.

H. Narrow Band (NB-IoT)

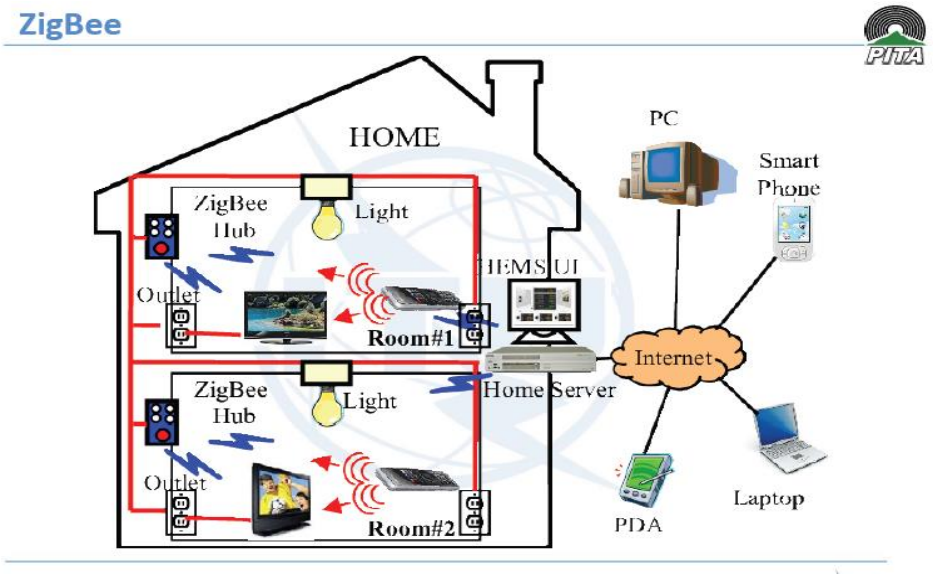
NB-IoT (Narrow-band IoT) is a network that belongs to the LTE-IoT type of wireless network. It is based on the operation of 4G mobile networks. The main features of this network are high coverage, lower power consumption than other LTE-based networks, and even cost, as NB-IoT is more affordable than eMTC. The coverage is large and ranges from 450 MHz to 3.5 GHz. According to Qualcomm, NB-IoT can work on both 2G and 3G networks. The total throughput of the NB-IoT network was reduced to 200 kHz. The NB-IoT network operates on the principle of half-duplex communication and uses TDD (Time Division Duplex) and FDD (Frequency Division Duplex). This divides the available band (into multiple frequencies or time bands) and the transmitter or receiver uses this allocated band to receive or send data.

I. ZigBee

The network originated in the early 21st century in the USA. It was standardized in 2003 by IEEE. ZigBee is a wireless network that has taken its place in the market by its specific features. While other wireless networks have tried for years to introduce new features and improve their performance, ZigBee has remained focused on 8bit microcontrollers. The most famous feature of this network is the long life of the device. The lifespan can be long up to 10 years or even more. ZigBee is similar to Bluetooth technology but has features such as low data rate, and therefore stands high with transfer reliability and good price. Although not used to transmit large amounts of data, some versions of the ZigBee protocol have been modified for such use. Due to its characteristics, it is often used for sensor networks and the transmission distance is up to 100 meters if there is good optical visibility



ZigBee topology



ZigBee home application

J. Cellular

3G/4G/LTE —Carrier broadband has come a long way in the last decade with huge improvements in data handling over mobile operator broadband networks, since the days in the early to late 2000s of 2G and GPRS. GPRS does however have IoT relevance, as it can be useful still in device interconnection in remote areas. However, with 2G networks being decommissioned it might not be around much longer. 3G/4G and LTE have broadband capabilities with high data throughput availability. It is 3G/4G and LTEs wide coverage footprint that makes it so enticing for IoT urban and rural applications. However, it has its drawbacks when used for device connectivity as it requires high power, so has poor battery life. Those failings can be mitigated by connecting many devices to a local 3G router, for example in a hub-spoke topology, which can make better use of the service available. However, because mobile networks are designed for voice and broadband data the price of a data plan and SIM for a remote router can be prohibitive. Mobile operators are beginning to offer IoT service plans so that might change, but even if the price does fall, the inherent design of the networks will not, and that will still make them technically inefficient for devices that communicate only tiny amounts, intermittently.